8.1 Similar Polygons

Geometry 8.1 · Student sheet

Name _____ Period ____ Date _____

I can…

- write a similarity statement, the congruent angles, and a statement of proportionality for similar polygons.
- use a scale factor to find a missing side length in similar polygons.
- find the perimeter and area of a polygon from a similar polygon.

Vocabulary · fill in the missing word

Similar figures — same shape, not necessarily the same size: _____ angles are congruent and corresponding side lengths are proportional

Scale factor — the ratio of _____ side lengths (image : original)

Statement of proportionality — an equation showing the ratios of all pairs of _____ sides are equal

Corresponding parts — parts in the same position, matched by the order of the letters in the _____ statement

Part 1 · Discovery: enlarge a triangle

On graph paper, draw right △ABC with legs 3 and 4 (the hypotenuse is 5). Draw it again with every side **2 times** as long, then **3 times** as long. Check the angles with a protractor.

Triangle	Side lengths	Perimeter	Area	Perimeter ÷ original	Area ÷ original
Original	3, 4, 5			1	1
× 2					
× 3					

What happened to the angles? What happened to the perimeter? To the area?

Part 2 · Notes and examples (fill in each note, then work the example)

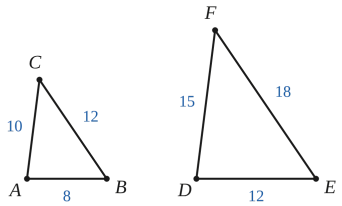
Similar polygons: Two polygons are **similar** when corresponding angles are _____ and corresponding side lengths are _____.

Scale factor: The ratio of corresponding side lengths is the _____ _____.

Corresponding parts: If △ABC ~ △DEF, then ∠A ≅ ∠____, and AB corresponds to _____.

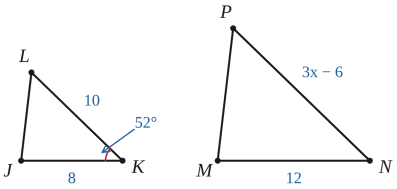
Example 1

△ABC ~ △DEF. Find the scale factor from △ABC to △DEF. List the pairs of congruent angles and write a statement of proportionality.



Example 2

△JKL ~ △MNP. JK = 8, KL = 10, MN = 12, and NP = 3x − 6. Find x. If m∠K = 52°, what is m∠N?



Perimeters of similar polygons: if the scale factor is k, the ratio of the perimeters is _____.

Example 3

△ABC ~ △DEF. AB = 9 and DE = 15. The perimeter of △ABC is 36. Find the perimeter of △DEF.

Areas of similar polygons: if the scale factor is k, the ratio of the areas is _____.

Example 4

Two triangles are similar. Their areas are 50 cm² and 18 cm². A side of the larger triangle is 15 cm. Find the corresponding side of the smaller triangle.

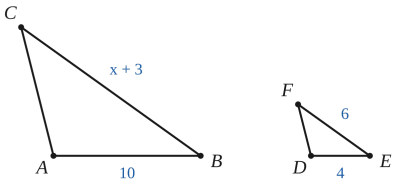
Example 5

△RST ~ △XYZ with RS = 12 and XY = 8. The altitude from T in △RST is 9. Find the altitude from Z in △XYZ.

Part 3 · Partner practice

Partner 1

△ABC ~ △DEF. AB = 10, BC = x + 3, DE = 4, and EF = 6. Find x.



Partner 2

Two similar pentagons have perimeters 20 in. and 30 in. The smaller pentagon has an area of 48 $in.^2$. Find the area of the larger pentagon.